

Nutrition by Design: Why the Future of School Meals Is Bigger Than One Ingredient



Karnataka's recent decision to expand eggs in school meals has reignited an old debate about nutrition.

Should schools serve eggs?

Should children who don't eat eggs receive an alternative?

Is one food better than another?

These are important conversations. But I can't help wondering if we're asking the wrong question.

Because the future of school nutrition will never be decided by one ingredient.

It will be decided by how intelligently we design every plate.



The search for a nutritional hero

Every few years, we seem to look for the next food that will solve malnutrition. Eggs. Milk. Fortified biscuits. Protein powders. Supplements.

Each enters the conversation with the hope that it will close nutritional gaps and improve children's health.

Many of these interventions have value. Eggs, for instance, provide high-quality protein and several essential nutrients.

Fortification has helped reduce specific micronutrient deficiencies in many settings.

But here's the challenge.

When we begin searching for a single nutritional hero, we risk reducing one of society's most complex challenges to one ingredient.

Malnutrition has never been the result of one missing food. It is shaped by dietary diversity, access to nutritious foods, affordability, maternal health, sanitation, education, food environments, agricultural systems, and poverty.

Complex problems rarely have single-solution answers.

Nutrition is no exception.

What if we asked a different question?

Instead of asking,

"Should we add eggs?"

or

"What should replace eggs?"

perhaps we should ask something far more useful.



How do we design meals that consistently deliver the nutrition every child deserves?

That shift changes the conversation completely.

Because nutrition isn't something we add.

It is something we design.

From ingredients to systems

Imagine building a bridge.

No engineer begins by asking,

"Should we use more steel?"

Instead, they ask,

"What structure will safely carry the load?"

Steel may be part of the answer.

Concrete may be another.

Each material has a role.

The strength comes from the design.

School meals deserve the same thinking.

Instead of building nutrition around one ingredient, we should build it around outcomes.

Does this meal provide sufficient protein?

Does it supply iron, calcium, zinc, folate and healthy fats?

Does it include enough fibre?

Will children actually enjoy eating it?

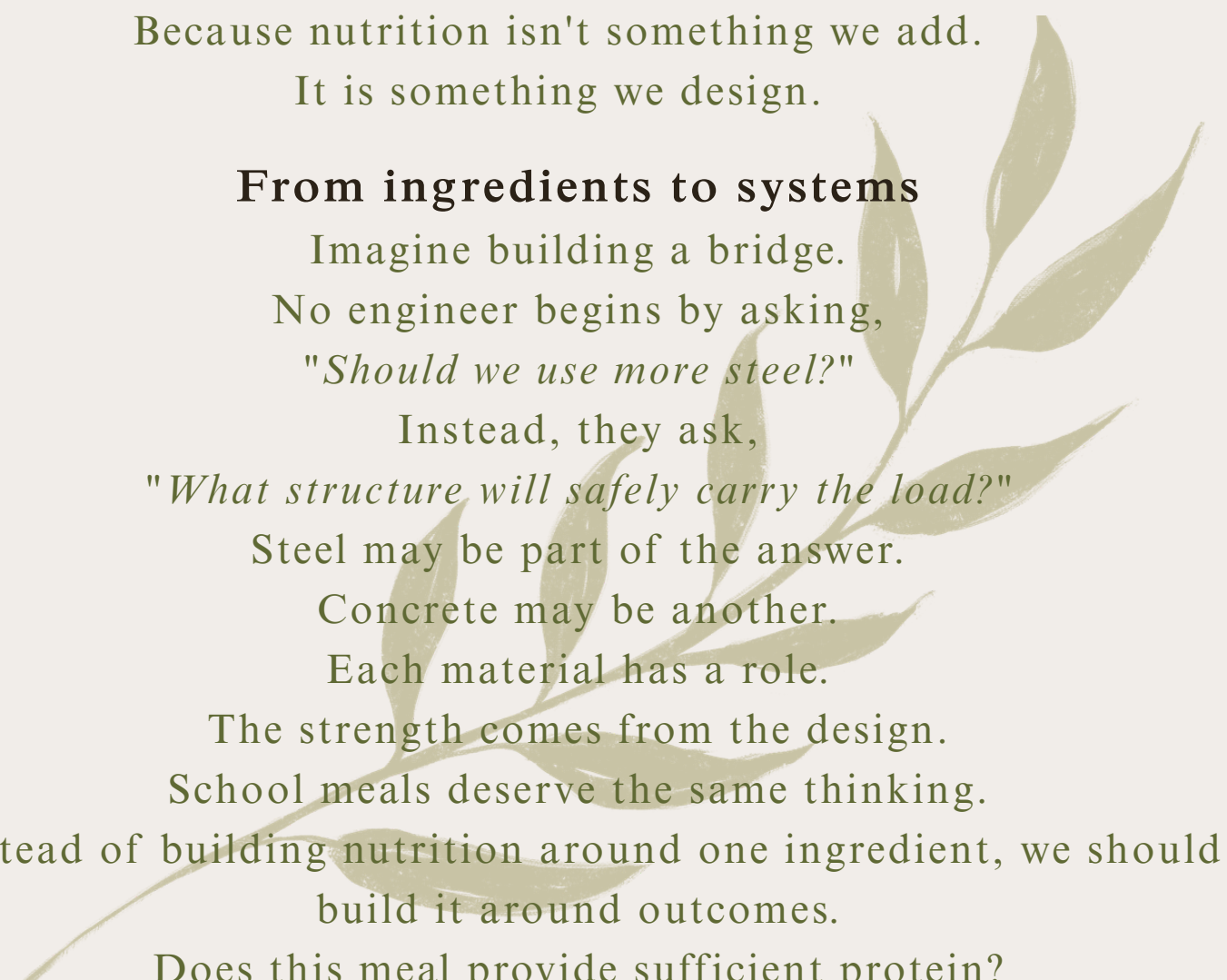
Can it be prepared consistently?

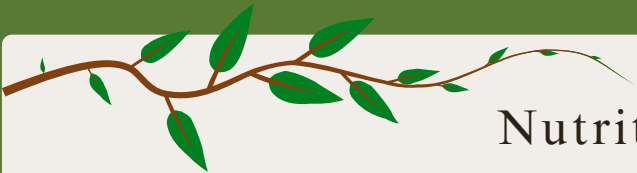
Can it be sourced locally?

Can families recognise and relate to these foods?

These are design questions.

And better questions usually lead to better solutions.





Nutrition by Design

I believe it's time we started thinking about Nutrition by Design.
A meal should never be judged simply by the presence or absence of
one ingredient.

It should be judged by whether it delivers complete nourishment.

That means considering:

Nutritional adequacy rather than individual foods.

Dietary diversity rather than dietary monotony.

Seasonal ingredients rather than standardised menus.

Regional food cultures rather than one-size-fits-all solutions.

Long-term health rather than short-term interventions.

This is where nutrition and food systems begin to meet.

Every plate shapes a food system

A school meal is never just a meal.

It is one of the largest public food procurement systems in the country.

Every ingredient on that plate creates demand somewhere.

Demand shapes what farmers grow.

What farmers grow influences biodiversity.

Biodiversity affects soil health, climate resilience and the stability of our food supply.

When schools regularly purchase diverse grains, pulses, vegetables and seasonal produce, they don't simply feed children.

They create markets for local farmers.

They help preserve traditional crops.

They encourage more resilient agricultural landscapes.

And they expose children to a wider diversity of foods during the years when lifelong eating habits are formed.





Nutrition policy, therefore, is also agricultural policy. It is environmental policy. It is education policy. It is economic policy.

Every plate tells a much bigger story than what's served for lunch.

Designing for nourishment, not debates

Whether a meal includes eggs or not should not be the only measure of its success.

The more important question is whether the overall meal consistently provides children with the nutrients they need in ways that are practical, culturally appropriate, economically viable and environmentally responsible.

A thoughtfully designed meal can support children's health while also strengthening local food systems.

Those objectives don't have to compete.

They can reinforce one another.

A new way to measure success

Perhaps it's time we judged school meals differently.

Not by asking,

"What's on today's menu?"

But by asking,

- Did this meal nourish the child?
- Did it encourage dietary diversity?
- Did it support local farmers?
- Did it strengthen biodiversity?
- Can it be sustained for decades—not just budget cycles?

When those questions guide our decisions, individual ingredients become part of a larger solution rather than the entire solution.



Beyond one ingredient

This isn't an argument against eggs.

Nor is it an argument in favour of replacing one food with another.

It's an argument against believing that any single ingredient can carry the weight of an entire nutrition strategy.

Children deserve more than isolated interventions.

They deserve meals that are intentionally designed to nourish growing bodies, support farming communities, celebrate regional food cultures and build resilient food systems.

Because the future of school nutrition isn't about choosing one ingredient over another.

It's about designing every plate with purpose.

Pooja Kothari | Founder The Nourished Roots

